
SENTINEL STACKS

Remediation Sentinel — Agent 3 Ship Report

Module: [remediate_findings_hardening_kit](#) — Tier 2, Dry-Run Default

Engine v1.1.0 · FIX_MAP 87 · TOOL_STANDARDS v1.0 · 2026-07-25

Repository: github.com/ibkballo/secops-pipeline-lab

Executive Summary

The **Remediation Sentinel** is a Tier-2 hardening kit engine that ingests scanner findings in TOOL_STANDARDS JSON format and produces **ready-to-review remediation artifacts**: Terraform templates for cloud infrastructure controls, nginx configuration fragments for perimeter defenses, and YAML runbooks for every finding. It operates exclusively in **dry-run mode**, packaging artifacts into ZIP kits — it never mutates live infrastructure.

This report documents the engine's architecture, FIX_MAP coverage across three security domains (AWS, Azure, Perimeter), full validation results against mock fixtures, compliance framework mapping (NIST 800-53, SOC 2, ISO 27001, GDPR), and the engine's programmatic contract.

Key Metrics

87	8	3/3
FIX_MAP Entries	Security Domains	Domains Validated
54/54	0	93
Findings Mapped	Unmapped	Artifacts/scan

Architecture

The engine implements a **universal shell** with a domain-pluggable FIX_MAP brain. Scanners emit findings; the engine matches each finding by `check_id` (exact key for AWS/AZ/GOV, title-pattern for VULN/IDENT/FIND), renders the appropriate template fragments via string substitution (not Python `str.format`, so Terraform braces pass through safely), and assembles a structured kit directory.

Scanner	check_id Pattern	Resolution
ai_infra_auditor_aws	AWS-001 ... AWS-018	Exact key
ai_infra_auditor_azure	AZ-001 ... AZ-018	Exact key
ai_governance_mapper	GOV-SSL ... GOV-REDIRECT (12)	Exact key
ai_vuln_hunter	VULN-001 ... VULN-014	Title pattern map
ai_identity_guard	IDENT-001 ... IDENT-012	Title pattern map
ai_network_auditor / api_scout / data_scout	FIND-001 ... FIND-009	Title pattern map

Kit Output Structure

Each kit directory contains:

- **terraform/** — .tf files for cloud controls (AWS, Azure only)
- **configs/** — nginx .conf fragments and security.txt (perimeter only)
- **runbooks/** — .yml per finding with steps, effort, reversibility
- **manifest.json** — kit metadata, findings index, dry_run flag
- **README.md** — human summary and apply instructions

Kits are zipped on dry_run=True and gitignored. Only the engine ships.

FIX_MAP Coverage

Domain	Controls	Terraform	Config Fragments	Runbook Only
AWS Infrastructure	18	10	—	8
Azure Infrastructure	18	11	—	7
GOV (Headers/Policy)	12	—	12	12
VULN (OWASP/Injection)	14	—	5	14
IDENT (Auth/JWT)	12	—	4	12
FIND (Data-Leak/Admin)	9	—	2	9
DATA (PII/Storage)	2	—	—	2
NET (Port Exposure)	2	—	—	2
TOTAL	87	21	23	87

Every finding receives a runbook YAML. Cloud findings also receive Terraform templates. Perimeter findings receive nginx/header configuration fragments where applicable.

Validation Results

All tests executed with dry_run=True against mock fixtures.

Test	Status	Total	Mapped	Unmapped	Artifacts
AWS Mock Vulnerable	PASS	18	18	0	35
Azure Mock Vulnerable	PASS	18	18	0	32
Perimeter Synthetic	PASS	12	12	0	26
AWS Clean Path	PASS	0	—	—	empty shell
Azure Clean Path	PASS	0	—	—	empty shell

Severity Distribution (Combined 54 Findings)

Critical	High	Medium	Low	Info
14	18	16	0	0

Compliance Framework Mapping

Every generated runbook and Terraform template references the applicable compliance controls. The engine maps findings to four frameworks:

Framework	Controls Referenced	Count
NIST 800-53 Rev 5	AC-2, AC-3, AC-6, AU-2, AU-3, AU-4, AU-6, AU-11, CM-2, CM-8, CP-9, IA-2, IA-5, IA-6, RA-5, SC-7, SC-8, SC-12, SC-13, SC-28, SI-4, SI-7	22
SOC 2 (Trust Services)	CC6.1, CC6.2, CC6.3, CC6.6, CC6.7, CC7.1, CC7.2, CC7.3, CC7.5	9
ISO 27001:2022	A.9.1.2, A.9.2.1–A.9.2.3, A.9.4.1–A.9.4.3, A.10.1.1–A.10.1.2, A.12.1.1, A.12.3.1, A.12.4.1, A.13.1.1, A.13.2.1, A.14.2.1, A.16.1.5, A.18.1.3	17
GDPR	Art. 25, Art. 30, Art. 32(1)(a)(b)(c)(d), Art. 33	6

Engine Contract

`run(params: dict) → dict` — returns a TOOL_STANDARDS-compliant JSON report with execution, summary, findings[], and metadata sections.

Parameter	Type	Default	Description
target	str	required	Path to scanner JSON report
output_dir	str	hardening_kits	Output directory for kit
dry_run	bool	True	Zip kit; never mutates infrastructure
fix_scope	list[str]	[]	Filter to specific check_ids
params	dict	{}	Pass-through for scanner compatibility

Response Summary Fields

Field	Type	Meaning
total_findings	int	Input findings count
fixes_mapped	int	Findings with matching FIX_MAP entries
fixes_unmapped	int	Findings without matching entries (needs FIX_MAP expansion)
artifacts_written	int	Total files generated (TF + conf + runbooks)
kits_generated	int	Number of kits created (always 1 or 0)
kit_path	str null	Path to generated ZIP
fix_map_size	int	Current FIX_MAP entry count

Ship Checklist

Item	Status
Engine compiles, no syntax warnings	✓
AWS 18/18 mapped, 0 unmapped	✓

Item	Status
Azure 18/18 mapped, 0 unmapped	✓
Perimeter 12/12 mapped, 0 unmapped	✓
Clean-path (no findings → skip)	✓
Terraform braces: single { } in output	✓
nginx regex: correct backslash escaping	✓
PowerShell UTF-8 BOM: engine handles BOM	✓
tool_registry.json at version 1.1.0	✓
dry_run always True	✓
GitHub commit	1381530

Known Limitations

- **No cloud mutation.** Engine emits artifacts only. Apply is manual or via separate automation.
- **Perimeter configs are examples.** nginx fragments use correct syntax but must be reviewed per deployment.
- **Title-based matching for VULN/IDENT/FIND.** If scanner title formats change, FIX_MAP needs updating.